

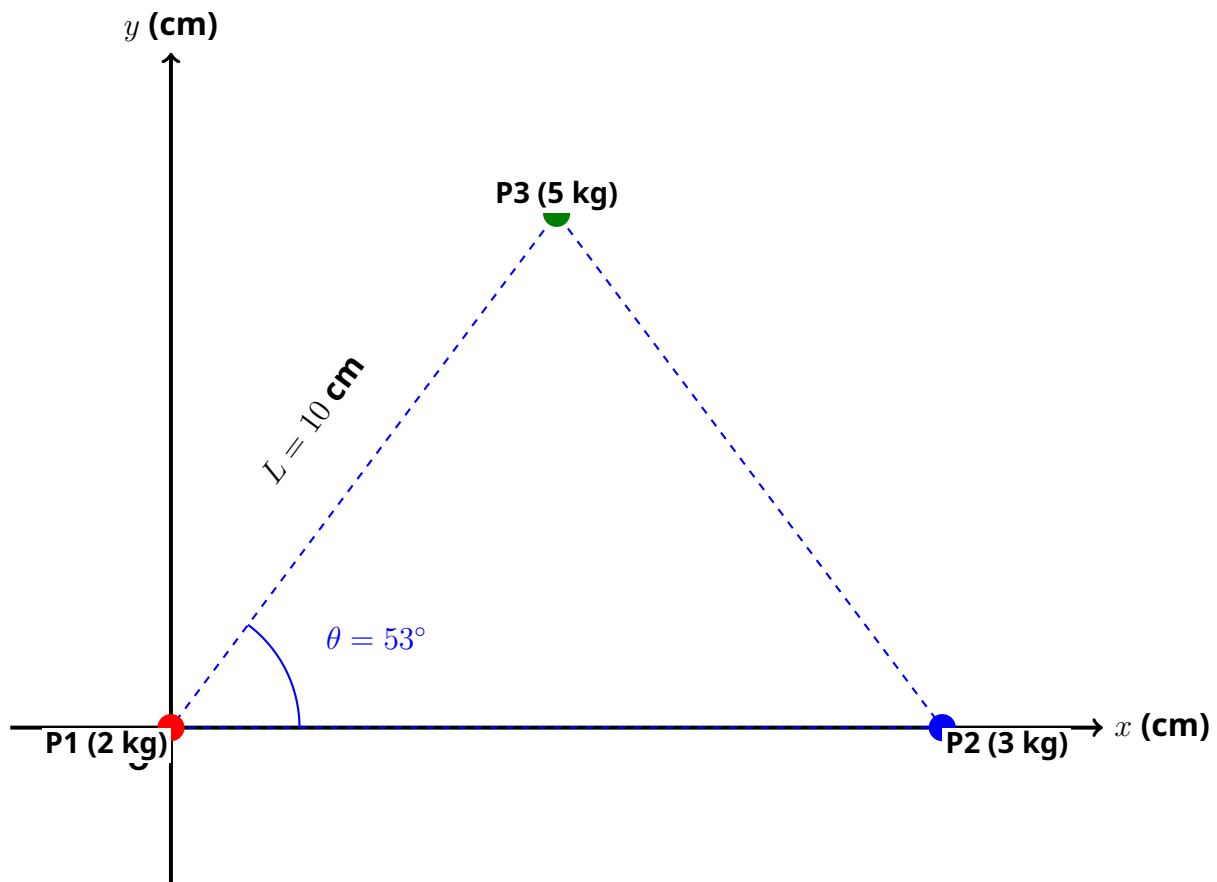
Assessment Problem 4: Geometric Layout

Problem Statement

Consider a system of **3 particles** located at the vertices of an isosceles triangle. The base of the triangle lies directly on the x -axis.

- **P1** ($m_1 = 2$ kg) is located at the left end of the base, coinciding with the origin $O(0, 0)$.
- **P2** ($m_2 = 3$ kg) is located at the right end of the base on the positive x -axis.
- **P3** ($m_3 = 5$ kg) is located at the main top vertex.

The two equal sides of the triangle each have a length of $L = 10$ cm. The base angle (the angle between the x -axis and the equal side) is $\theta = 53^\circ$. (Given for calculations: $\cos(53^\circ) = 0.6$ and $\sin(53^\circ) = 0.8$)



Questions

Please solve on a separate sheet of paper.

1. **(1.5 points)** Use sine and cosine to calculate the exact coordinates (x, y) of particle **P3**. Then, deduce the coordinates of **P2**.
2. **(0.5 points)** Calculate the **Total Mass** (M_{sys}) of the system.
3. **(3 points)** Determine the **coordinates** (x_G, y_G) of the Center of Mass. Show your work.